1. DESIGN STANDARD

1.1 THE DYNAMIC SMOKE EXTRACTION SYSTEM FOR THIS PREMISE IS DESIGNED ACCORDING TO THE CODES OF PRACTICE FOR MINIMUM FIRE SERVICES INSTALLATIONS AND EQUIPMENT – APRIL 2012, ISSUED BY FIRE SERVICES DEPARTMENT, HKSAR. (HEREAFTER REFERRED TO AS "FSI-COP").

1.2 FSD CIRCULAR LETTER NO. 4/96 PART IV SMOKE EXTRACTION SYSTEM (HEREAFTER REFERRED TO AS "CIRCULAR LETTER 4/96").

2. DESIGN CRITERIA

2.1 THE DYNAMIC SMOKE EXTRACTION SYSTEM SHALL BE DESIGNED IN ACCORDANCE WITH FSI-COP.

2.2 ALL SYSTEMS SHALL COMPRISE OF EXTRACTION AND MAKE-UP AIR INSTALLATIONS.

2.3 THE EXTRACTION RATE SHALL BE EQUIVALENT TO NOT LESS THAN EIGHT AIR CHANGES PER HOUR OF THE TOTAL COMPARTMENT VOLUME. THE DESIGN VOLUME IS TAKEN TO BE 7000 CUBIC METRES FOR B1 LOADING & UNLOADING AREA & B2 LOADING & UNLOADING AREA DUE TO ACTUAL VOLUME OF LESS THAN 7000 CUBIC METRES.

2.4 THE MAKE-UP AIR FLOW SHALL BE EQUIVALENT OR NOT LESS THAN THE 80% OF THE EXTRACTION RATE.

2.5 SEPARATE SYSTEMS SHALL BE PROVIDED FOR EACH COMPARTMENT AND EACH SYSTEM SHALL COMPRISE AT LEAST TWO INDEPENDENT PLANTS AND DUCTWORK I.E. FOR A PROPORTION OF THE AREA AND EXTRACT/SUPPLY VOLUMES.

2.6 SMOKE SHALL NOT TRAVEL MORE THAN 30 METER BEFORE ENTERING THE NEAREST POINT OF INLET TO THE EXTRACT SYSTEM AND AT LEAST ONE EXTRACT POINT SHALL BE PROVIDED WITHIN EACH 500 SQUARE METRES UNIT OF FLOOR AREA.

2.7 POINT(S) OF SMOKE EXTRACTION SHALL BE FROM HIGH LEVEL IN THE SPACE CONCERNED AND SHALL BE REASONABLY DISTRIBUTED.

2.8 MAKE-UP AIR SHALL ENTER AT A LOW LEVEL AND/OR IN SUCH A MANNER AS TO AVOID PREMATURE MIXING WITH THE HOT GAS AND SHORT CIRCUITING TO THE EXTRACTION POINTS..

2.9 MAXIMUM VELOCITIES, BASED ON FREE AREA OF THE GRILLE, SHALL BE AT
1) AT MAKE-UP AIR INLETS WHERE MECHANICALLY PROPELLED (6M/s);
2) AT EXTRACT GRILLES OR OUTLETS (6M/s)

2.10 TO PREVENT RECIRCULATION OF SMOKE AS FAR AS POSSIBLE, DISCHARGE OUTLETS FOR SMOKE SHALL BE SEPARATED BY NOT LESS THAN 5 M IN ANY DIRECTION FROM ALL AIR INLETS OR OTHER OPENINGS INTO ANY BUILDING. THEY SHALL NOT DISCHARGE INTO ANY MEANS OF ESCAPE NOR A FREE AIR FIREMAN'S STAIRCASE. NO DISCHARGES SHALL BE AT A HEIGHT ABOVE THE SURROUNDING HORIZONTAL SURFACE OF LESS THAN 3M TO THE BOTTOM OF THE OUTLET AND WHERE BELOW 6M SHALL NOT DISCHARGE DOWNWARDS. NO DISCHARGES SHALL BE UNDER ANY CANOPY OR OVERHANG.

2.11 DUCTWORK AND LOUVER USED FOR "NORMAL" PURPOSES MAY BE UTILIZED FOR SMOKE EXTRACTION PURPOSES PROVIDED THAT UNDER SMOKE EXTRACTION MODE THE CONSTRUCTION AND OPERATION OF THE SYSTEM COMPLIES WITH THE REQUIREMENTS HEREIN.

2.12 EQUIPMENT HANDLING SMOKE SHALL BE SUITABLE FOR CONTINUOUS OPERATION AT 250°C FOR NOT LESS THAN 1 HOUR AS PER FSD REQUIREMENT. THIS INCLUDES FANS, MOTORS, DAMPERS, DRIVES AND DUCTWORK.

2.13 LOCAL MOTOR CONTROL PANEL (LMCP) SHALL BE PROVIDED WITH FRR 120/120/120 WITH SELF-CLOSING DOOR TO SUIT FSD REQUIREMENT.

2.14 THE SMOKE EXTRACTION AND MAKE-UP AIR DUCTWORKS SHALL BE OF ROBUST DESIGN TO WITHSTAND THE MECHANICAL IMPACT AS PER FSD REQUIREMENTS. THE SMOKE EXTRACTION AND MAKE-UP AIR DUCTWORKS SHALL BE CONSTRUCTED OF FIRE RATED MATERIALS AND/OR PROTECTED BY FIRE RATED ENCLOSURES. THIS FIRE RESISTING DUCTWORK SHALL BE CONSTRUCTED FROM MATERIAL WHICH, APART FROM ITS FIRE RESISTING QUALITY, SHALL BE CAPABLE OF RESISTING ACCIDENTAL MECHANICAL DAMAGE. TO THIS END, THE DUCTWORK SHALL PASS THE HARD BODY IMPACT TEST IN ACCORDANCE WITH BS EN 1128

2.15 ALL SYSTEMS SHALL BE "FAIL-SAFE" TO ENSURE A FREE PASSAGE OF SMOKE.

2.16 MOTORIZED FIRE AND SMOKE DAMPER (MFSD) SHALL COMPLY WITH UL555S

2.17 IF SMOKE EXTRACT/MAKEUP AIR FANS ARE INSTALLED WITHIN THE SERVED COMPARTMENT, THE SYSTEM INCLUDING FANS, MOTOR, DRIVES, ELECTRICAL WORK, DUCTWORK LINKING FANS AND THE BOUNDARY OF THE COMPARTMENT ETC. SHOULD BE PROTECTED BY A FIRE RESISTING MATERIAL OF RATING NOT LESS THAN 1 HOUR.

2.18 EMERGENCY ELECTRICAL SUPPLIES SHALL BE CAPABLE OF OPERATING SIMULTANEOUSLY ALL SYSTEMS RELEVANT TO THE SMOKE EXTRACTION SYSTEM IN THE TWO ADJACENT COMPARTMENTS HAVING THE HIGHEST TOTAL ELECTRICAL LOAD. "ADJACENT" SHALL BE IN ANY DIRECTION.

2.19 FIRE RESISTANCE RATING OF ALL FIRE RATED ENCLOSURE/ DUCTWORK SHALL COMPLY WITH FSI-COP 2012 AND COP FOR FTRE SAFETY IN BUILNING 2011

3. SYSTEM DESCRIPTION

TOTAL THREE (3) SETS OF DYNAMIC SMOKE EXTRACTION SYSTEMS DESIGNED IN ACCORDANCE WITH THE PRESCRIPTIVE CODE REQUIREMENT SHALL BE PROVIDED TO SERVE THE BASEMENTS WHICH ARE SHOWN IN THE AIR FLOW SUMMARY LISTED BELOW.

4. SYSTEM CONTROL AND ACTUATION

4.1 THE FAN SYSTEM OPERATION AND OPEN/CLOSE STATUS OF DAMPERS UNDER "NORMAL", "FIRE" AND "NO POWER" MODES ARE FULLY DESCRIBE AS FOLLOWS AND IN THE MODE TABLE. (DRAWING NO.2503373A/SES/102)

A. NORMAL MODE
THE SYSTEM SHALL REMAIN INACTIVE WITH "OFF" SIGNAL AS INDICATED ON THE CONTROL PANEL AT G/F

B. FIRE MODE
THE SMOKE EXTRACTION SYSTEM WILL BE ACTIVATED AUTOMATICALLY UNTIL MANUAL RESET UPON RECEIPT OF FOLLOWING SIGNAL:

a) ACTUATION OF ANY 2 NUMBERS OF SMOKE DETECTORS OF THE SERVED COMPARTMENT.

THE FOLLOWING COMPONENTS/EQUIPMENT OF THE SMOKE EXTRACTION SYSTEM SERVING THE COMPARTMENT SHALL BE ACTUATED.

a) THE CORRESPONDING SMOKE EXTRACTION/MAKEUP AIR FANS SERVING THE COMPARTMENT.
b) ALL MOTORIZED FIRE AND SMOKE DAMPER (MFSD) IN THE COMPARTMENT ACCORDING TO MODE TABLE IN DRAWING NO. 2503373A/SES/102.

MAKEUP AIR FAN(MAF) SHALL BE INTERLOCKED WITH SMOKE EXTRACTION FAN(SEF) ACCORDING TO THE FOLLOWING SCENARIOS:
a) FOR EACH SYSTEM, IF ANY ONE SEF SERVING THE COMPARTMENT IS "ON", THE CORRESPONDING MAKEUP AIR FAN (MAF) SHALL BE STARTED UP.
b) FOR EACH SYSTEM, IF ANY ONE SEF SERVING THE COMPARTMENT IS "OFF" OR "FAILURE", THE CORRESPONDING MAF SHALL BE SHUT DOWN AND CANNOT BE STARTED UP.
c) FOR EACH SYSTEM, IF ANY ONE MAF SERVING THE COMPARTMENT IS "OFF" OR FAILURE", THE SEF WILL BE CONTINUED TO OPERATE.

THE DYNAMIC SMOKE EXTRACTION SYSTEM SHALL BE ABLE FOR MANUALLY RESET WITH INDICATION IN SMOKE EXTRACTION SUPERVISORY CONTROL PANEL. MANUAL OVERRIDE SWITCH SHALL BE PROVIDED IN THE SMOKE EXTRACTION SYSTEM SUPERVISORY CONTROL PANEL AT G/F FIRE CONTROL CENTER.

UPON ACTUATING THE MANUAL OVERRIDE DEVICE, THE SYSTEM STATUS SHALL BE KEPT UNCHANGED.

ALL VENTILATION AND AIR CONDITIONING SYSTEMS SERVING THE SAME PLANT ROOM/ COMPARTMENT SHALL BE SHUT DOWN AUTOMATICALLY UPON ACTUATION OF THE CORRESPONDING SMOKE EXTRACTION SYSTEM.

C. NO POWER MODE
FAILURE OF ONE OR MORE PHASES OF THE MAINS SUPPLY SHALL INITIATE AUTOMATIC STARTING OF THE ESSENTIAL POWER SUPPLY.

FULL LOAD TRANSFER SHALL TAKE PLACE AUTOMATICALLY. ESSENTIAL POWER SYSTEM IS PROVIDED AS AN ALTERNATIVE SOURCE OF POWER IF THE NORMAL POWER SUPPLY MODE IS FAILED.

IF THE ESSENTIAL POWER SYSTEM IS ALSO FAILURE, THE FANS WILL NOT BE IN OPERATION AND THE ASSOCIATED DAMPERS WILL BE RETURNED TO FAIL-SAFE POSITION.

5. ELECTRICAL POWER SUPPLY

5.1 EMERGENCY ELECTRICAL POWER SUPPLY SHALL BE PROVIDED FOR THE SMOKE EXTRACTION SYSTEM INCLUDING MAKE-UP SYSTEM.

5.2 FIRE RESISTANT CABLE AS APPROVED BY FSD SHALL BE USED FOR BOTH POWER SUPPLY AND CONTROL CABLES.FIRE RESISTANT CABLE SHALL COMPLY WITH FSD CIRCULAR LETTER NO. 2/2017

6. DUCT CONSTRUCTION

6.1 DUCT CONSTRUCTION SHALL BE, AS A MINIMUM REQUIREMENT, TO NOT LESS A STANDARD THAN HEATING AND VENTILATING CONTRACTORS' ASSOCIATION SPECIFICATION DW144 AND ITS SUBSEQUENT AMENDMENTS.

6.2 ALL DUCTWORK INCLUDING BUILDERS WORK DUCTS OR SHAFTS AND OTHER CONSTRUCTION, IN THESE SYSTEMS, SHALL, WHERE OUTSIDE THE SERVICED FIRE COMPARTMENT, BE PRESSURE TESTED TO HEATING AND VENTILATION CONTRACTORS' ASSOCIATION SPECIFICATION DW143 AND ITS SUBSEQUENT AMENDMENTS.

6.3 FIRE RESISTING DUCTWORK SHALL BE CONSTRUCTED FROM MATERIAL WHICH, APART FROM ITS FIRE RESISTING QUALITY, SHALL BE CAPABLE OF RESISTING ACCIDENTAL MECHANICAL DAMAGE. TO THIS END, THE DUCTWORK SHALL PASS THE HARD BODY IMPACT TEST IN ACCORDANCE WITH BSEN 1128 WITH THE WEIGHT BEING DROPPED THROUGH NOT LESS THAN ONE METER.

6.4 THE DUCTWORKS SERVED FOR THE DYNAMIC SMOKE EXTRACTION SYSTEM (INCLUDE SMOKE EXTRACTION AND MAKEUP AIR SYSTEM) SHALL BE FIRE RATED DUCT (FRR 240/240/240) IN COMPLIANCE WITH BS476 PART 24.

AIR FLOW SUMMARY

SYSTEM	FLOOR	SERVED AREA	AREA (m^2)	HEIGHT (m)	VOLUME (m^3)	DESIGN VOLUME (m^3)	MINIMUM REQUIRED SMOKE EXTRACTION RATE (m^3/s)	PROVIDED REQUIRED SMOKE EXTRACTION RATE (m^3/s)	MINIMUM REQUIRED MAKEUP AIR RATE (m^3/s)	PROVIDED REQUIRED MAKEUP AIR RATE (m^3/s)	EQUIPMENT DESIGNATION IN DRAWING	FUNCTION	INSTALLATION LOCATION	MOUNTING	MINIMUM FAN AIR FLOW RATE (m^3/s)
1	B2	LOADING & UNLOADING AREA	B 725.711 (A/GBP/35)	5.20	B 3773.697	7000.0	15.556	15.60	12.48	12.48	SEF-B2-01	SMOKE EXTRACTION	B2 COMMERCIAL CARPARK	CEILING MOUNT	7.80
											SEF-B2-02	SMOKE EXTRACTION		CEILING MOUNT	7.80
											MAF-B2-01	MAKEUP AIR SUPPLY	B2 LOADING & UNLOADING AREA	CEILING MOUNT	6.24
											MAF-B1-03	MAKEUP AIR SUPPLY	B1 LOADING & UNLOADING AREA	CEILING MOUNT	6.24
2	B1	LOADING & UNLOADING AREA	599.422 (A/GBP/35)	6.20	3716.416	7000.0	15.556	15.60	12.48	12.48	SEF-B1-03	SMOKE EXTRACTION	B1 COMMERCIAL CARPARK	CEILING MOUNT	7.80
											SEF-B1-04	SMOKE EXTRACTION		CEILING MOUNT	7.80
											MAF-B1-05	MAKEUP AIR SUPPLY	B1 LOADING & UNLOADING AREA	CEILING MOUNT	6.24
											MAF-B1-04	MAKEUP AIR SUPPLY		CEILING MOUNT	6.24
3	B1	B RETAIL AREA (A)	267.112 (NOTE 2)	5.87 (NOTE 3)	1567.947 B	B 10878.424 B	B 24.174 B	24.30	19.44	19.44	SEF-B1-01	SMOKE EXTRACTION	B1 SMOKE EXTRACTION FAN ROOM (FOR COMMERCIAL)	FLOOR MOUNT	12.15
		RETAIL AREA (B)	401.728 (NOTE 2)	5.60 (NOTE 3)	2249.677						SEF-B1-02	SMOKE EXTRACTION	B1 SHOP B01	CEILING MOUNT	12.15
		B RETAIL AREA (C)	1412.160 (NOTE 2)	5.00 (NOTE 3)	7060.800 B						MAF-B1-01	MAKEUP AIR SUPPLY	B1 MAKE UP FAN ROOM	FLOOR MOUNT	9.72
											MAF-B1-02	MAKEUP AIR SUPPLY	B1 SHOP B01	CEILING MOUNT	9.72

NOTES:
1. THE DESIGN VOLUME SHALL BE CONSIDERED TO BE 7000 CUBIC METER (m^3) FOR ANY COMPARTMENT OF 7000 CUBIC METER (m^3) OR LESS.
2. RETAIL AREA IS B 2081.000 m^2 (REFER TO DRAWING NO. A/GBP/35). (RETAIL AREA (A) 267.112 + RETAIL AREA (B) 401.728 + RETAIL AREA (C) 1412.160 = 2081.000 m^2) B (MEASURED FROM GBP LAYOUT).
3. HEIGHT REFER TO DRAWING NO. 2503373A-SES-003 FLOOR TO FLOOR ELEVATION. B

FIRE SERVICE INSTALLATION PLANS AS CERTIFIED BY THE AUTHORIZED PERSON AND FIRE SERVICE INSTALLATION CONTRACTOR / CONSULTANT
Date 08 SEP 2026
Signed (FUNG Kam-wah)
for Director of Fire Services

LOO Kwan Wai
(RPE No.: RP0260258)

NO.	DATE	REVISIONS	DRAFTED	CHECKED	DESIGNED	REVIEWED
B	06/2026	FSD THIRD SUBMISSION	HYH	WWC	HYH	TKL
A	09/2025	FSD SECOND SUBMISSION	HYH	WWC	HYH	TKL
0	05/2025	FSD FIRST SUBMISSION	HYH	WWC	HYH	TKL

CLIENT:
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Central, Hong Kong

PROJECT ARCHITECT & AUTHORIZED PERSON:
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呂元祥建築師事務所
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STRUCTURAL/GEOTECHNICAL
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Kowloon Bay, Hong Kong

BUILDING SERVICES ENGINEER:
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Kowloon Bay, Hong Kong

QUANTITY SURVEYOR:
ARCADIS
17/F, Two Harbour Square, 180 Wai Yip Street,
Kwun Tong, Kowloon
Hong Kong

FIRE SERVICES CONTRACTOR:
開創工程有限公司
GRAND CREATIVE ENGINEERS LIMITED

PROJECT:
PROPOSED DEVELOPMENT AT BAILEY STREET & WING KWONG STREET, TO KWA WAN, KOWLOON
INLAND LOT NO. 11279
(URA PROJECT KC-009)

TITLE:
SMOKE EXTRACTION SYSTEM-
GENERAL NOTES AND
AIR FLOW SUMMARY

DATE: 06/2026 SCALE: NTS@A0

DRAFTED:	CHECKED:	DESIGNED:	REVIEWED:
HYH	WWC	HYH	TKL

JOB NO.	DRAWING NO.	REV. NO.
2503373A	2503373A/SES/002	B